RENDER_TEST_SUITE.md — Comprehensive Test Scenarios

> **Version:** 1.0.0
> **Coverage:** All RENDER_RULES.md rules
> **Automation:** `pytest tests/` or `python -m pytest tests/ -v`

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§1 Layout Governance Tests

1.1 Overflow Prevention (RULE-010, RULE-011, RULE-012)

Test ID	Description	Input	Expected Result
LAY-001	Line exceeding 200 chars triggers warning	250-char line in .md	WARNING: no_overflow
LAY-002	`overflow: hidden` CSS is blocked	CSS with overflow:hidden	ERROR: no_overflow
LAY-003	Normal content passes without findings	Standard 80-char lines	No findings
LAY-004	Figure below 60% scale triggers error	Figure with min_scale violation	ERROR: scale_minimum

1.2 Spacing (RULE-020, RULE-021, RULE-022)

Test ID	Description	Input	Expected Result
LAY-010	Content block gap ≥16px	Layout with 12px gap	ERROR: minimum_gap
LAY-011	Section break ≥32px	Section with 24px break	ERROR: section_break
LAY-012	Valid spacing passes	Compliant layout	No findings

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§2 Typography Tests

2.1 Heading Hierarchy (RULE-030, RULE-031, RULE-032)

Test ID	Description	Input	Expected Result
TYP-001	Sequential headings pass	H1 → H2 → H3	No findings
TYP-002	Skipped heading level detected	H1 → H3 (skipping H2)	ERROR: stable_heading_hierarchy
TYP-003	Multiple H1s flagged	Two H1 headings	INFO: multiple_h1
TYP-004	Orphan heading detected	H2 with no following content	WARNING

2.2 Font Size Constraints (RULE-040–RULE-052)

Test ID	Description	Input	Expected Result
TYP-010	Body text ≥16px screen	14px body text	ERROR: min_body_size
TYP-011	Heading ≤48px screen	56px heading	ERROR: max_heading_size
TYP-012	Font ≥12px at all viewports	10px at mobile	ERROR: min_font_size

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§3 Contrast Tests

3.1 WCAG AA Compliance (RULE-060, RULE-061, RULE-062)

Test ID	Description	Input	Expected Result
CTR-001	Black on white passes (21:1)	fg:#000 bg:#FFF	PASS (ratio: 21:1)
CTR-002	Light gray on white fails	fg:#CCC bg:#FFF	FAIL (ratio: ~1.6:1)
CTR-003	Dark gray on white passes	fg:#333 bg:#FFF	PASS (ratio: ~12.6:1)
CTR-004	Large text at 3:1 passes	24px fg:#666 bg:#FFF	PASS (ratio: ~5.7:1)
CTR-005	Theme light mode validation	SEMANT-IC_THEME.yaml light	All pairs ≥4.5:1
CTR-006	Theme dark mode validation	SEMANT-IC_THEME.yaml dark	All pairs ≥4.5:1
CTR-007	High contrast mode validation	SEMANT-IC_THEME.yaml high_contrast	All pairs ≥7:1

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§4 Content Integrity Tests

4.1 Orphan Bullets (RULE-080, RULE-081)

Test ID	Description	Input	Expected Result
CNT-001	Single bullet item flagged	`- Only item`	WARNING: no_orphan_bullets
CNT-002	Two+ bullet items pass	`- Item 1\n- Item 2`	No findings
CNT-003	Single numbered item flagged	`1. Only item`	WARNING: no_orphan_bullets
CNT-004	Nested list depth >3 flagged	4-level nested list	WARNING: max_nesting

4.2 Figure References (RULE-090, RULE-091)

Test ID	Description	Input	Expected Result
CNT-010	Defined and referenced figure passes	Fig with ref in text	No findings
CNT-011	Defined but unreferenced figure	Fig without text ref	WARNING
CNT-012	Referenced but undefined figure	Text ref without fig	ERROR

4.3 Speaker Notes (RULE-100)

Test ID	Description	Input	Expected Result
CNT-020	Notes ≥50 chars passes	60-char speaker notes	No findings
CNT-021	Notes <50 chars flagged	30-char speaker notes	WARNING
CNT-022	Missing notes on slide file	No speaker_notes field	WARNING

4.4 Semantic Cards (RULE-101, RULE-102)

Test ID	Description	Input	Expected Result
CNT-030	Complete semantic card passes	slide_id + purpose + audience	No findings
CNT-031	Missing slide_id	purpose + audience only	ERROR
CNT-032	Missing purpose	slide_id + audience only	ERROR
CNT-033	No front matter on slide	Slide file without YAML	ERROR

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§5 Slide ID Tests

5.1 ID Format (RULE-110, RULE-111)

Test ID	Description	Input	Expected Result
SID-001	Valid ID passes	arch-overview-001	VALID
SID-002	Missing sequence fails	arch-overview	INVALID
SID-003	Non-numeric sequence fails	arch-overview-abc	INVALID
SID-004	Empty string fails	”“	INVALID
SID-005	Special characters fail	arch/overview-001	INVALID
SID-006	Single segment fails	architecture	INVALID

5.2 Uniqueness (RULE-111)

Test ID	Description	Input	Expected Result
SID-010	Unique IDs pass	All distinct IDs	No violations
SID-011	Duplicate IDs flagged	Same ID in two files	ERROR: duplicate

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§6 Lineage Tests

6.1 Verification Hook (Input_Master/ Processing)

Test ID	Description	Input	Expected Result
LIN-001	Binary file creates .mock	test.pptx in Input_Master/	.mock sidecar created
LIN-002	SHA256 hash computed	Any binary file	Valid 64-char hex hash
LIN-003	Manifest updated	New file processed	lineage_manifest.json updated
LIN-004	Duplicate file detected	Same file re-dropped	Existing entry updated
LIN-005	Invalid file rejected	Corrupted binary	Error logged, no .mock

6.2 Manifest Integrity

Test ID	Description	Input	Expected Result
LIN-010	Valid manifest schema	Correct JSON structure	Schema validation passes
LIN-011	Missing required field	Record without sha256	Schema validation fails
LIN-012	Invalid timestamp format	Non-ISO-8601 date	Schema validation fails

§7 Integration Tests

7.1 End-to-End Pipeline

Test ID	Description	Steps	Expected Result
INT-001	Full asset ingestion	Drop binary → verify → check manifest	Complete lineage chain
INT-002	Linter + Enforcer combo	Lint then enforce slide IDs	Both pass on valid content
INT-003	Theme contrast full check	Load theme → check all pairs	All AA requirements met

7.2 CI/CD Workflow

Test ID	Description	Trigger	Expected Result
INT-010	PR governance check	PR to main	All validators run
INT-011	Asset verification on push	Push to Input_Master/	Verification hook executes
INT-012	Lint failure blocks merge	PR with linter errors	Check status: failure

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§8 Test Execution

```
# Run all unit tests
python -m pytest tests/ -v

# Run specific test module
python -m pytest tests/test_render_linter.py -v

# Run with coverage
python -m pytest tests/ --cov=engines --cov-report=html

# Run integration tests only
python -m pytest tests/ -v -m integration

# Run linter directly
python engines/RENDER_LINTER.py docs/

# Run contrast checker
python engines/WCAG_CONTRAST_CHECKER.py --theme engines/SEMANTIC_THEME.yaml

# Run slide ID enforcer
python engines/SLIDE_ID_ENFORCER.py docs/

# Run verification hook
python engines/verification_hook.py --input-dir Input_Master/ --data-dir _data/
```